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# Vantage® *NXT* Research Ultrasound Systems

## FastSave Tutorial: L22-14vX FlashAngles

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# 1. Tutorial Overview

This tutorial shows how to modify a L22-14vX FlashAngles sequence to use FastSave for continuous RF streaming while retaining live monitoring with reduced reconstruction rates.

# 2. Profile the existing script

```
matlab

    SetUpL22_14vXFlashAngles
    vsProfile start
    VSX
```

Run the script for ~60sec to collect enough statistics, then stop and exit the sequence.

```
vsProfile stop[resultTable, results, benchmarks] = vsProfile( 'review');

>> results(2).MedianDuration

ans =

24.4675
```

This result says that it takes about 24.5ms to reconstruct 1 frame of 7 plane waves compounded.

```
>> Receive(1).endSample*na*128*2

ans =

 4358144
```

- Median reconstruction time: ~24.5 ms per compounded frame (7 angles).
- DMA size per sub-frame: 4,358,144 bytes (≈ 4.16 MiB / 4.36 MB). This relatively small transfer size will not fully saturate the DMA capabilities of the hardware; grouping sub-frames into larger super-frames is recommended.

# 3. Hardware frame timing and constraints

timeToNextAcq (between angle acquisitions) = 160us

timeToNextAcq (between frames) = 19040us

This original time between frames is set for a hardware framerate of 50fps. This can be reduced to a minimum value of the time between angle acquisitions.

- Therefore, one compounded frame (7 angles) duration: 7 × 160 µs = 1,120 µs ≈ 1.12 ms, i.e., ~893 frames/s

**Note**: Reconstruction (≈24.5 ms) is much slower than the raw hardware acquisition rate. To provide live monitoring without dropping frames, we must group multiple sub-frames into a super-frame for DMA transfer and perform reconstruction at a lower effective frame rate.

# 4. Choosing the number of sub-frames per super-frame

Constraints and calculations:

- Reconstruction time must be less than the super-frame acquisition time. With a reconstruction time of 24.5 ms and a single-frame acquisition time of ~1.12 ms:Number of sub-frames >= 24.5 ms / 1.12 ms ≈ 21.9 ⇒ choose ≥ 23 sub-frames.
    - Super-frame acquisition time (23 sub-frames) = 1.12 ms × 23 ≈ 25.8 ms → ≈ 38.8 super-frames/s.
- DMA transfer efficiency: empirically, larger DMA transfers (≥ 100 MB) achieve higher sustained DMA rates on this platform. Using the per-sub-frame sizeExample (25 sub-frames): 25 × 4,358,144 = 108,953,600 bytes ≈ 103.9 MiB (108.95 MB).
    - Per sub-frame: 4,358,144 bytes (~4.16 MiB / 4.36 MB).
    - Minimum sub-frames for ~100 MB: 100,000,000 / 4,358,144 ≈ 22.9 ⇒ choose ≥ 23 (rounded up to an integer).
- Upper bound from per-file limits: ensure the super-frame size remains below any 32-bit/implementation limits (e.g., 2 GB):Max sub-frames (≈2,000 MB / 4.36 MB) ≈ 458 ⇒ safely above the operating range.

Recommendation for this example: choose 50 sub-frames for higher DMA efficiency while maintaining a usable live reconstruction rate.

- Super-frame DMA size (50 sub-frames): 50 × 4,358,144 = 217,907,200 bytes ≈ 207.7 MiB (217.9 MB).
- Super-frame acquisition time: 1.12 ms × 50 ≈ 56 ms ⇒ effective live reconstruction rate ≈ 17.9 fps (≈18 fps).
- Minimum DMA transfer duration (approx): 0.218 GB / 7 GB/s ≈ 31 ms (order-of-magnitude estimate); this is consistent with the observed acquisition timing for a sustained transfer.

These calculations show that 25 sub-frames is close to the minimum acceptable value; 50 sub-frames is a reasonable compromise for higher DMA efficiency and an ~18 fps live reconstruction rate.

## 5. Modify the sequence script (high-level steps)

Reduce the inter-frame wait (set timeToNextAcq between frames to the per-angle interval so frames are captured continuously). For example, update the SeqControl argument used to set the frame interval (replace the previous 20 ms-derived argument with 160):
Below are the minimal, focused edits to adapt the example sequence. Preserve existing sequence semantics where possible and test incrementally.

```
matlab

%% before
SeqControl(3).argument = 20000 - (na-1)*160;  % example prior value (20 ms)

%% after (set inter-angle / inter-frame spacing to 160 us)
SeqControl(3).argument = 160;  % microseconds
```

Set the chosen number of sub-frames per super-frame (example uses 50):

```
matlab

P.nSubFrames = 50;
```

Configure FastSave variables (base filename + target drives):

```
matlab

fsFileName = 'testFileName';
if isunix()
    saveMethod = 'fastsaveLinux';
    fsDriveNames =
{'/media/verasonics/WD1','/media/verasonics/WD2','/media/verasonics/WD3','/media/ve
rasonics/WD4'};
elseif ispc()
    saveMethod = 'fastsaveWindows';
    fsDriveNames = {'D:\','E:\','F:\','G:\'};
end
```

Create the FastSave external process entry:

```
matlab

fastSaveProcess = 2; % choose an available process index
Process(fastSaveProcess).classname = 'External';
Process(fastSaveProcess).method = saveMethod;
Process(fastSaveProcess).Parameters = { ...
    'srcbuffer','receive', ...
    'srcbufnum', 1, ...
    'srcframenum', -1, ... % process the last frame in the receive buffer
    'dstbuffer','none' ...
};
```

Add external save events and required SeqControls to transfer and mark DMA transfers:

```
matlab

% Example snippet (integrate into the Event creation loop):
Event(n).info = 'Transfer To Host';
Event(n).tx = 0; Event(n).rcv = 0; Event(n).recon = 0; Event(n).process = 0;
Event(n).seqControl = [nsc, nsc + 1];
SeqControl(nsc).command = 'transferToHost';
SeqControl(nsc+1).command = 'waitForTransferComplete';
SeqControl(nsc+1).argument = nsc; % wait index
nsc = nsc + 2;
n = n + 1;

% Save-event marker for grouping sub-frames (markTransferProcessed)
saveEvents = [saveEvents n];
Event(n).info = 'save all subframes';
Event(n).tx = 0; Event(n).rcv = 0; Event(n).recon = 0; Event(n).process = 0;
Event(n).seqControl = nsc;
SeqControl(nsc).command = 'markTransferProcessed';
SeqControl(nsc).argument = nsc - 2;
nsc = nsc + 1;
n = n + 1;
```

Create the super-frame receive structure and populate per-subframe attributes. The original structure is preserved; ensure `Receive` is sized for `na * P.nSubFrames * Resource.RcvBuffer(1).numFrames` entries and that `framenum`/`acqNum` are assigned consistently. Example:

```
matlab

maxAcqLength = ceil(sqrt(P.endDepth^2 + ((Trans.numelements-1)*Trans.spacing)^2));
Receive = repmat(struct('Apod', ones(1,Trans.numelements), ...
    'startDepth', P.startDepth, ...
    'endDepth', maxAcqLength, ...
    'TGC', 1, 'bufnum', 1, 'framenum', 1, 'acqNum', 1, ...
    'sampleMode', 'NS200BW', 'mode', 0, 'demodFrequency', 17.8571, ...
    'rcvFilter', 1, 'callMediaFunc', 0), 1, na * P.nSubFrames *
Resource.RcvBuffer(1).numFrames);

rcvI = 1;
for frameI = 1:Resource.RcvBuffer(1).numFrames
    Receive(rcvI).callMediaFunc = 1;
    acqI = 1;
    for subFrameI = 1:P.nSubFrames
        for angleI = 1:na
            Receive(rcvI).framenum = frameI;
            Receive(rcvI).acqNum = acqI;
            acqI = acqI + 1;
            rcvI = rcvI + 1;
        end
    end
end
```

Modify the Event loop to iterate over frames → sub-frames → angles:

```
matlab

for frameI = 1:Resource.RcvBuffer(1).numFrames
    acqI = 1;
    for subFrameI = 1:P.nSubFrames
        for angleI = 1:na
            Event(n).info = 'Full aperture.';
            Event(n).tx = angleI;
            Event(n).rcv = acqI;
            Event(n).recon = 0;
            Event(n).process = 0;
            Event(n).seqControl = 2;
            n = n + 1;
            acqI = acqI + 1;
        end
    end
end
```

Increase the receive buffer rows per frame to accommodate all sub-frames:

```
matlab

Resource.RcvBuffer(1).rowsPerFrame = P.nSubFrames * na * 6144; % adjust as needed
for max depth
```

Add a GUI control to start/stop saving (toggle button) and a callback that enables/disables the external process for the saved events.

```
matlab

    import vsv.seq.uicontrol.VsToggleButtonControl

    UI(3).Control = VsToggleButtonControl('LocationCode','UserB1',...
        'Label','Start Fast Save',...
        'Callback',@saveAcquisition);

    % **** Callback routines used by UI Controls. ****
    function saveAcquisition(~,~,UIValue)
    saveEvents = evalin('base', 'saveEvents');
    Event = evalin('base', 'Event');
    for i = 1:length(saveEvents)
        Event(saveEvents(i)).process = UIValue*2;  %for external
    end
    assignin('base','Event',Event);
    Control = evalin('base','Control');
    if isempty(Control(1).Command), n=1; else, n=length(Control)+1; end
    Control(n).Command = 'update&Run';
    Control(n).Parameters = {'Event'};
    assignin('base','Control', Control);
    end
```

(Optional) Enable acquisition time tags for timing validation:

```
matlab

Resource.VDAS.acqTimeTagEnable = true;
```

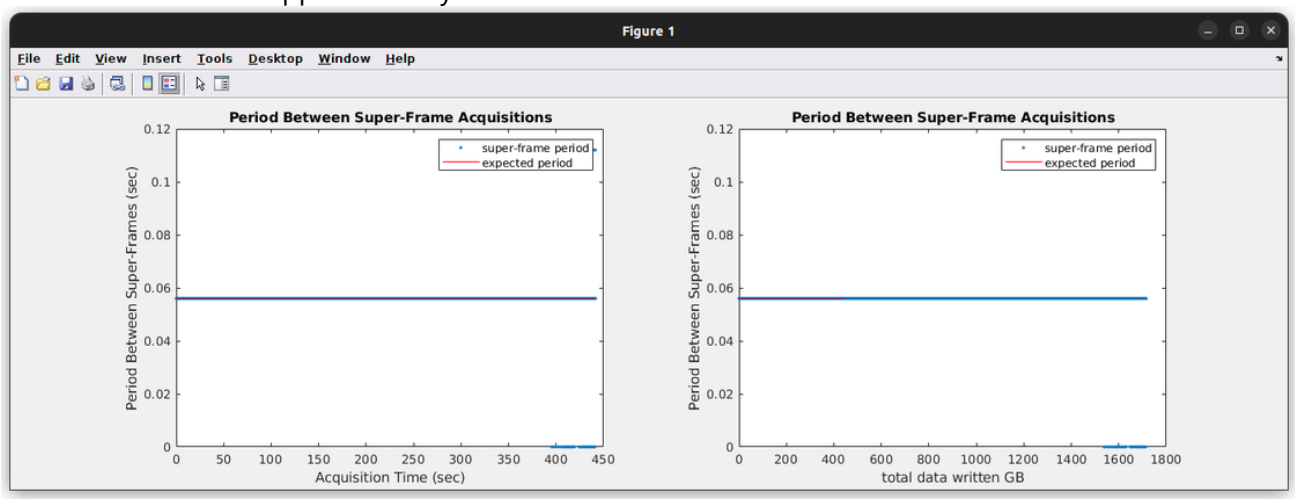
## 6. Test and validate

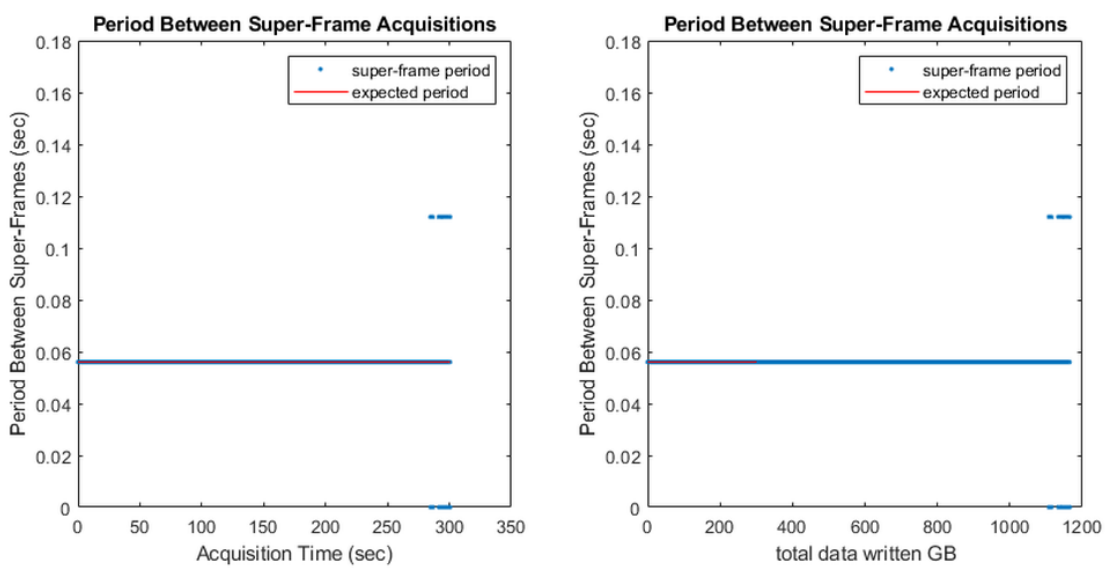
Use or adapt a test runner (`demoRun.m`) to:

1. Verify storage mount points and formatting.
2. Optionally trim/prepare the drives and warn before any destructive wipe.
3. Launch the sequence with profiling enabled.

Once you feel like you have collected enough data:

1. Disable the file streaming with the same GUI button clicked for launching.
2. Run the script: `validateCollectedData`.
3. After a run, validate with `validateCollectedData`.
   The validation plot shows period between frames vs time and vs total storage written; spurious deviations indicate dropped or delayed frames.



## 7. Example summary statistics

Total # files captured: 4530 files

Total duration of acquisition: 254 seconds

Total size of collected data: 0.99 TB

Percent of Storage Used: 25

Average Data Throughput: 3.9 GB/s

With the settings recommended (50 sub-frames per super-frame), the sequence can provide live reconstruction of a 7-angle plane-wave compounding image at ~18 fps while streaming continuously to disk for up to ~17 minutes (assuming the four storage devices sustain ≥ 1 GB/s each after caches fill).

## 8. Recommendations and notes

- Test incrementally: first verify acquisition timing and buffer sizing without FastSave enabled, then enable FastSave and validate transfers.
- Keep the fio benchmark and filesystem tuning guidelines from the main documentation in mind — filesystem choice and mount options materially affect sustained throughput.
- Shut down unnecessary background processes and disable indexing / antivirus on dedicated acquisition drives during tests.
- Record exact parameters (P.nSubFrames, SeqControl arguments, Receive sizes, filesystem, fio job file) alongside the test results for reproducibility.

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